

Meeting Preparation Document

Panda Benchmark Investigation

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Purpose of this document. This covers only what the experiment log and reading report do not give you. Read this *after* reading both of those. It contains: key numbers to know without looking them up, anticipated questions with prepared answers, the narrative of each negative result, and what to ask for from the meeting.

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1 Key Numbers to Know Cold

Do not look these up during the presentation. These are the numbers most likely to be cited or challenged.

1.1 Weather advantage – the anchor result

H	Panda MAE	Chronos MAE	Advantage	p	n
96	0.638	0.812	+0.174	< 0.001	20
192	0.722	0.958	+0.236	0.001	20
336	0.848	1.084	+0.236	< 0.001	20

All three horizons are the same order of magnitude. The result is not a fluke of one horizon.

1.2 ETTh1 and ETTh2 – no reliable advantage

ETTh1: no significant advantage at any horizon (all $p > 0.6$). Direction of advantage is inconsistent across horizons.

ETTh2: one significant horizon ($H = 336$, $p = 0.013$), three others not significant and inconsistent in direction.

Honest summary: **neither ETT dataset shows reliable Panda advantage.**

1.3 Heterogeneity bottleneck – strongest mechanistic finding

Subset	Het	Panda MAE	Chronos MAE	Adv	p
Homo matched (H=96)	0.086	0.331	0.700	+0.369	0.004
Hetero controlled (H=96)	0.853	0.605	0.866	+0.261	0.004
Homo matched (H=336)	0.086	0.841	1.147	+0.305	0.004
Hetero controlled (H=336)	0.853	0.867	1.175	+0.309	0.012

Panda MAE nearly doubles ($0.331 \rightarrow 0.605$) from homogeneous to heterogeneous at matched difficulty and matched difficulty CV. Chronos shows no monotone trend across heterogeneity levels.

1.4 Component ablation numbers

- **Channel attention (Weather, H=96):** univariate Panda MAE 0.554 vs multivariate 0.611. Univariate is better. Not significant ($p = 0.074$, $n = 8$).
- **Channel attention (Burgers $\nu = 1.0$):** univariate advantage +0.003 (significant, $p = 0.004$); multivariate advantage -0.002 (not significant). Univariate wins.
- **Head fine-tuning (H=96):** base MAE 0.609, FT MAE 0.699. Fine-tuning hurts by 0.091 ($\approx 15\%$ relative increase).
- **Harmonic oscillator:** Panda advantage +0.370, $p = 0.004$.
- **Van der Pol:** Panda advantage +0.011, $p = 0.027$.

1.5 Decomposition – revised

Experiment 20 (Chronos residual ablation): Chronos also degrades by +0.277 after FFT decomposition ($p = 0.016$, $H = 96$). Magnitude is comparable to Panda degradation in earlier decomposition experiments. Earlier interpretation (Panda specifically depends on full signal) must be revised.

1.6 Topology

Correlation dimension estimates (Grassberger-Procaccia, **unreliable estimator**):

Dataset	Corr. dim	Panda wins?
Weather	0.889	Yes
Lorenz	0.862	Yes
Burgers	0.741	Yes
ETTh1	1.541	No
ETTh2	1.617	No

Pattern is suggestive. Estimator is unreliable on real multivariate data. Treat as pilot measurement only, not a finding.

1.7 Evaluation protocol validity

Window independence was verified: all experiments use `np.linspace` over `max_start`, producing strides substantially larger than context length (512) across all datasets. Consecutive windows do not overlap. Wilcoxon signed-rank test independence assumption is satisfied.

2 Anticipated Questions and Prepared Answers

Q: Why does the Harmonic oscillator have such a large advantage?

We do not know. Three competing explanations, none ruled out:

1. Chronos specifically fails on pure sinusoidal inputs due to its discrete tokenisation scheme. The advantage reflects Chronos weakness rather than Panda strength.
2. Panda’s Koopman embedding represents sinusoids efficiently because sinusoidal functions are eigenfunctions of the Koopman operator for linear systems.
3. The result may be seed- or initial-condition-specific. One seed, $n = 8$ windows. Not verified for robustness.

What would distinguish these: run Chronos alone on Harmonic vs Van der Pol vs Duffing. If Chronos specifically collapses on Harmonic, explanation 1 is strongly supported.

Do not say: “Panda’s Koopman embedding is well-suited to periodic systems.” That is low-confidence speculation.

Q: Why does Van der Pol have such a small advantage?

Also unknown. Van der Pol is a limit cycle – bounded, periodic in the long run, but nonlinear with harmonics. Possible explanations:

1. Chronos handles limit cycle dynamics adequately because limit cycles produce quasi-periodic patterns that appear in real-world seasonal data.
2. Van der Pol at the parameters used may have a spectral structure that both models handle similarly.

The non-monotone pattern (Harmonic large, VanderPol small, chaotic large) may reflect spectral complexity rather than dynamical complexity. This is a hypothesis.

Q: Did you control for difficulty in the heterogeneity experiment?

Yes, twice.

First control: difficulty-matched subsets. Per-channel Chronos univariate MAE matched across homo/mixed/hetero (mean difficulty ≈ 0.80 – 0.84). Panda MAE still increases monotonically with heterogeneity (0.331, 0.475, 0.618 at H=96). Chronos stays flat.

Second control: coefficient of variation of difficulty matched. The `hetero_controlled` subset has difficulty CV = 0.110, comparable to homogeneous subsets. Panda MAE still 0.605 at H=96 vs 0.331 for homo matched. Both significant.

The heterogeneity effect on Panda survives both controls. Chronos shows no corresponding effect in either.

Q: Did you try more fine-tuning steps for the head?

No. 50 gradient steps is the only thing tested. However, 50 steps with ground-truth context-forecast pairs on the target domain should show at least some positive signal if the prediction head were the bottleneck. A head that is truly the bottleneck should improve immediately when given the right targets. The degradation at H=96 is 0.091 MAE in the wrong direction – a substantial movement. This is more consistent with encoder-head mismatch than insufficient training.

If asked: acknowledge honestly. “We cannot rule out that more steps would help, but the direction and magnitude of degradation at 50 steps suggests the bottleneck is upstream of the head.”

Q: What about the decomposition result – doesn’t it show Panda needs the periodic component?

This was the interpretation after Experiments 11 and 17. Experiment 20 (Chronos residual ablation) revised it. Chronos also degrades by +0.277 after decomposition, comparable in magnitude to Panda degradation. Since Chronos has no structural reason to depend on the periodic component, the most parsimonious explanation is that FFT decomposition introduces projection error that degrades both models. The result does not cleanly support Panda-specific signal dependence.

Do not say: “Panda’s advantage comes from the periodic component.” That was pre-Experiment 20 and has been revised.

Q: What is the positive mechanism behind Panda’s Weather advantage?

Unknown. Channel attention and prediction head have been ruled out as the driver. The remaining untested candidates are the Koopman lifting (fixed random polynomial and Fourier features) and the temporal attention and non-causal encoder. Both require code-level access to ablate.

Say this directly: “We have ruled out channel attention and prediction head. The Koopman lifting and temporal encoder are the remaining candidates. Testing these requires modifying the model architecture, which requires code access.”

Q: Is the topology result meaningful?

Treat it as suggestive only. The Grassberger-Procaccia estimator has known reliability problems on multivariate real-world data with limited samples. The pattern (Weather $\approx$ Lorenz $\approx$ Burgers $<$ ETTh by correlation dimension) is interesting because it clusters exactly along Panda-wins vs Panda-loses. But a single noisy estimate per dataset is not evidence.

What it motivates: persistent homology (Vietoris-Rips filtration) would give Betti numbers and persistence diagrams – quantities with established theoretical guarantees. Do not present the correlation dimension numbers as findings. Present them as a pilot measurement that justifies a more rigorous analysis.

Q: Why didn’t you test temporal attention?

It requires removing it from the architecture. All experiments in this investigation were inference-time ablations or minimal fine-tuning on the existing checkpoint. Temporal attention cannot be ablated at inference time without changing the model.

If asked what you would do with compute: priority order is (1) Koopman lifting ablation, (2) temporal attention ablation, (3) proper node embedding implementation.

Q: Why is Panda not better on ETTh at any horizon?

Two non-exclusive explanations:

1. Pretraining distribution mismatch. ETTh channels are quasi-linear with complex seasonal patterns. Correlation dimension estimates (1.54, 1.62) are substantially higher than training distribution attractors.
2. Chronos is specifically well-suited to ETTh-type data – it was trained on a large corpus of real-world time series including electricity and oil-price patterns similar to ETT.

These two explanations are not distinguishable from our experiments.

Q: What would you run next if you had GPU compute?

Have a clear priority order:

1. **Koopman lifting ablation** (most discriminating): replace fixed random features with a learned linear projection of equal dimension. Evaluate on Weather, ETTh1,

Burgers. If advantage collapses, lifting is the positive mechanism. First experiment that could give causal evidence.

2. **Proper node embedding implementation** (most actionable): learned vector $\mathbf{e}_i \in \mathbb{R}^d$ per sensor injected into channel attention. Test whether the homo-to-hetero MAE gap shrinks.
3. **XXLTraffic evaluation**: test whether the heterogeneity pattern generalises to a domain where sensor heterogeneity is much more severe.

Q: Did you read the Panda paper’s own ablations?

Yes. The paper’s ablation (Section 5.2) shows channel attention is the largest single contributor on held-out chaotic systems. This appears to contradict our univariate ablation result on Weather and Burgers. The resolution: the paper’s ablation is on chaotic ODE systems where channel coupling is real and strong. Our ablation is on Weather (heterogeneous sensors) and non-chaotic Burgers (diffusion-dominated PCA modes). Channel attention is beneficial when channels have genuine dynamical coupling; it adds noise when they do not. Both results are correct in their respective domains.

Q: What does the Wilcoxon test actually test here?

One-sided Wilcoxon signed-rank test on per-window MAE differences (Panda better than Chronos). With $n = 8$ windows, the minimum achievable p-value is 0.004, which means all 8 windows showed Panda better. It carries no additional precision information beyond that. With $n = 20$, the floor drops to approximately 0.0001 and the test becomes genuinely informative. Window independence was verified: linspace-generated starts give strides larger than context length throughout.

Q: Why MAE instead of sMAPE?

sMAPE breaks down on instance-normalised data because values near zero make the denominator vanish, producing wild artifacts. Lorenz at $\rho = 15$ in the periodic regime showed Chronos sMAPE near zero – not genuine performance, a metric artifact. MAE is stable on normalised data. Within-dataset Panda vs Chronos comparisons are unaffected by scale sensitivity because both models see the same normalised windows.

3 The Narrative of Each Negative Result

Flora’s group may interpret negative results as “nothing worked.” Have a precise statement ready for each.

Univariate ablation (Exp 9, 27)

Rules out channel attention as the *driver* of Panda’s advantage on Weather and Burgers. Does not rule out that the temporal architecture (Koopman + non-causal encoder) is the driver. Does not say “Panda’s architecture is no better than univariate” – Panda univariate still beats Chronos significantly. The paper’s own ablation finding (channel attention helps on chaotic ODE systems) remains valid in its own domain.

Head fine-tuning (Exp 23)

Rules out lightweight head-only adaptation as a practical intervention. Does not rule out

joint fine-tuning of encoder and head together. The result points toward encoder-head co-adaptation as the architecture’s internal constraint.

Permutation entropy (Exp 21)

Rules out PE at order 3 as a useful complexity discriminator for this problem – it cannot discriminate periodic from chaotic Lorenz, and operates in incomparable numerical regimes across data types. Does not rule out complexity as a relevant factor. Higher-order PE, sample entropy, or topological invariants might succeed.

Decomposition experiments (Exp 11, 17, revised by Exp 20)

These do not establish Panda-specific signal dependence. Both models degrade comparably after decomposition. The experiments reveal the difficulty of cleanly isolating the periodic component, not a structural property of Panda.

Node embedding proxy (Exp 22)

Rules out that a scalar per-channel offset is sufficient to encode sensor identity. Does not rule out that a proper learned embedding vector would help. The directional effect at H=96 (hurts homogeneous, marginally helps heterogeneous) is weakly consistent with the G-SWaN direction.

4 Mapping to Flora’s Suggestions

Suggestion	What our experiments show	Status
G-SWaN / node embeddings	Heterogeneity bottleneck established through 3 controlled experiments. Scalar offset proxy directionally correct but insufficient.	Low–Medium confidence. Failure mode established; intervention untested properly.
XXLTraffic evaluation	Heterogeneous sensor bottleneck is the correct frame. Traffic sensors are the extreme case of Weather heterogeneity.	Well-motivated evaluation target. Not yet run.
Topological flow matching	Correlation dimension pilot suggestive: Weather and Lorenz cluster together, ETTh separates. Estimator unreliable. Persistent homology needed.	Low confidence. Hypothesis only; evidence quality insufficient to act on.
TimesNet-style pretraining	Both models equally hurt by decomposition (Exp 20). Panda loses to Chronos on periodic-only context (Exp 18). No evidence periodic handling is a bottleneck.	Deprioritised. Weakest current motivation.

5 Framing and What to Ask For

5.1 Opening framing

Do not start with “Panda fails on non-chaotic data.” That was the pre-presentation conclusion and it was wrong – Weather is non-chaotic and Panda wins substantially. Start with: “We tested whether Panda generalises beyond its chaotic ODE training distribution and found a more nuanced picture than expected.”

5.2 One-sentence research summary

If Flora asks for the bottom line at the end:

“Panda generalises to Weather and Burgers beyond its chaotic ODE distribution, does not generalise reliably to ETTh, and sensor heterogeneity is the most controlled failure mode identified – but the positive mechanism behind the advantage is not yet established.”

5.3 Handling uncertainty publicly

When you do not know the answer, use the epistemic label explicitly: “We observed X. We have competing hypotheses for why, but no experiment currently distinguishes them.” Flora will respect this more than a confident but unsupported narrative.

5.4 Two asks from this meeting

Both are directly tied to specific findings, which makes them much easier to justify.

1. **Code access to Panda’s architecture** for the Koopman lifting ablation. This is the one experiment that could give the first causal evidence for any architectural component. Replace fixed random features with a learned linear projection, evaluate on Weather and Burgers. If advantage collapses, lifting is the mechanism.
2. **Compute for proper node embedding implementation** on the homo/hetero Weather subsets. Not the scalar proxy – a learned embedding vector per sensor injected into channel attention. Test whether the homo-to-hetero MAE gap decreases.

5.5 What to say when you do not know

For the Harmonic oscillator anomaly and the Van der Pol minimum: “We have no causal explanation. These are the competing hypotheses, and this experiment would distinguish them.” Saying this cleanly is better than a forced narrative.

6 Things That Could Come Up Not in the Log

“Is Panda’s channel attention actually permutation equivariant?”

Yes, by design. No positional encoding is used in channel attention. Channels are processed as an unordered set. This is what makes it transfer from 3-channel training to arbitrary channel counts at test time – it never learned “there are exactly 3 channels,” it learned

“attend between channels based on content similarity.” This same property is what makes it insensitive to sensor identity.

“How does the Burgers result relate to the heterogeneity story?”

Burgers channels are PCA modes of the same field – homogeneous by construction. Panda wins on Burgers even at non-chaotic viscosity. This is *consistent* with the heterogeneity story (homogeneous channels are where Panda should do well) but adds no new evidence for it. The scientific value of the Burgers non-chaotic result is purely as a counterexample to the chaos-specific hypothesis – not as evidence for or against heterogeneity.

“Why is Panda better on Weather but not ETTh if both are non-chaotic?”

The correlation dimension estimates suggest Weather’s joint state-space trajectory may be geometrically closer to Panda’s chaotic training distribution than ETTh’s is. Weather has twenty-one channels with genuine physical coupling (pressure, wind, temperature, humidity); the joint dynamics may be more complex and nonlinear than ETTh’s seven channels of electricity and oil-related variables. However this explanation is speculative – the topology result is unreliable. Two other non-exclusive explanations: Chronos is specifically well-calibrated for ETTh-type seasonal patterns (having seen similar data in training), and ETTh’s channel count (7) gives Panda’s attention less cross-channel structure to work with.

“What is the Burgers advantage size relative to Weather?”

Weather advantage is +0.174 to +0.236 at matched difficulty, $n=20$, highly significant. Burgers advantage at statistically significant viscosity values ranges from +0.038 to +0.149, $n=8$ (minimum p floor). Burgers advantages are comparable in magnitude but less well-powered. The Burgers result is scientifically important not for the size of the advantage but for ruling out chaos as a prerequisite.